

Team Tracks & Task Checklists (M1–M5)

How to use this: each teammate picks the track that fits them best (niche hints below). Within a track, do the tasks **in the order listed**. Tasks marked ★ **unblock other people**, so finish them first and post in the group when they're done. Every task names its dependencies. "—" means you can start right away.

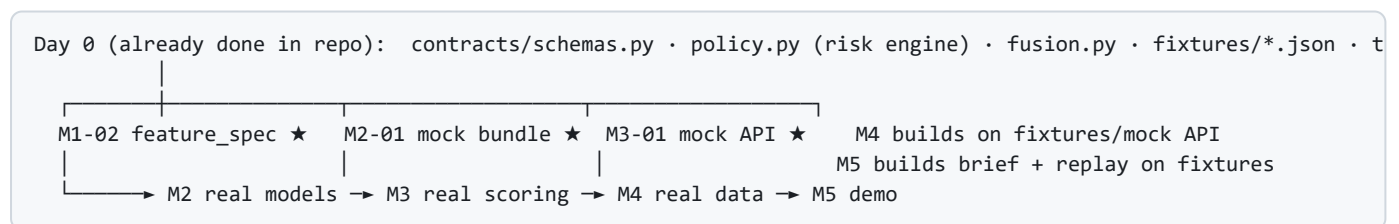
Each track has a GitHub issue with the same checklist (label `track:M1` ... `track:M5`). Tick boxes there as you go.

Datasets: corrected **CSE-CIC-IDS2018** = train / validate / test · **LUFlow** = real-traffic showcase with its own bundle. See [03 §3.0](#).

Pick your track

Track	Name	Pick this if you like...	Owns
M1	Data & Features	pandas, data cleaning, statistics	<code>ml/data/</code> (incl. dataset adapters), <code>nscore/features/</code> , <code>nscore/drift/</code>
M2	ML Modeling & MLOps	scikit-learn, experiments, Azure ML	<code>ml/train/</code> <code>evaluate/</code> <code>explain/</code> <code>registry/</code> , <code>nscore/detection/</code> , <code>nscore/bundle/</code>
M3	Backend & Platform	APIs, databases, DevOps	<code>api/</code> , <code>infra/</code> , <code>.github/workflows/</code>
M4	SOC Console (Frontend)	UI/UX, visualisation	<code>dashboard/</code>
M5	Security, GenAI & Demo	security, LLM prompting, storytelling	<code>api/app/services/brief.py</code> , <code>replay/</code> , risk weights in <code>nscore/contracts/policy.py</code> , the deck

Why the tracks can run in parallel



Only **four** hard hand-offs exist: `feature_spec.json` (M1→all), mock bundle (M2→M3), mock API (M3→M4), real bundle v1 (M2→M3). Everything else is built against the contracts and fixtures.

Integration milestones (everyone)

Milestone	Exit criteria	Owners
I0 Foundation	<code>feature_spec.json</code> merged · mock bundle loads in the API · mock API serves all endpoints · dashboard renders every page from mock data · CI green	M1, M2, M3, M4
I1 First real model	2018 bundle v1 loaded from a local ref · <code>replay</code> → <code>/v1/flows</code> → risk-scored incident appears in the dashboard	all
I2 Feature complete	Azure ML pull + cache fallback · briefs (LLM + template) · drift live · holdout-botnet bundle · LUFlow + LUFlow-recal bundles served · LOAO and real-world panels in the UI	all

Milestone	Exit criteria	Owners
I3 Demo ready	2 timed dry runs · recorded fallback video · model card + README final · deck claims checked against what's built	all

Schedule I3 at least **2–3 days before Round 2**, not the night before.

M1 — Data & Features

Status: all M1 tasks done on branch `track/M1` (reports in `docs/data_profile.md`, data section in `docs/model_card.md`, replay files in `replay/samples/`). The EDA notebook (M1-04) is the generated `docs/data_profile.md`, so every figure is reproducible by script.

- ☐ **M1-01** Download **corrected CSE-CIC-IDS2018** (distrinet-research.be/CNS2022; start this first, it's large) and **LUFlow** (github.com/ruzzzzzz/LUFlow or Kaggle). **Confirm the 2018 files contain** `Src IP`, `Dst IP`, `Timestamp`. Record source URLs, checksums and row counts per label in `data/README.md`. *Deps: —*
- ☐ **M1-02 ★** Write the canonical `nscore/contracts/feature_spec.json` (CIC schema): raw → snake_case names, dtypes, clip ranges, log1p flags, and the dropped columns with reasons (identifier or leakage). Run correlation pruning ($|p| > 0.95$) on a 2018 train sample. Open a PR; M2 and M3 review it. *Deps: M1-01 → unblocks M2, M3*
- ☐ **M1-03** `ml/data/adapters/cic2018.py`: process **one day file at a time** → canonical frame (meta + features + `family` + `period`), with Inf/NaN report (counts logged, nothing silently zero-filled), dedupe, label → `AttackFamily` mapping ("Attempted" → BENIGN), float32 parquet in `data/processed/`. *Deps: M1-01*
- ☐ **M1-04** EDA notebook + `docs/data_profile.md`: class counts per day, missing values, top correlated pairs, what got dropped and why. **Confirm the low-support decision for WebAttack** (283 flows in total, see `data/README.md`) and the measured facts: FTP-Patator is all `Attempted`, `SlowHTTPTest` is absent, Infiltration is 99.7% internal Nmap scanning. *Deps: M1-03*
- ☐ **M1-05** `ml/data/split.py`: **time-blocked split per (day, label), 70/15/15, with a 60 s purge at block boundaries**, saved as index files plus `split_manifest.json`. A fixed-seed **training sample** (all attack flows capped per class + a few million benign). A LOAO fold generator (one fold per family) and a **HOIC tool-holdout** fold. Unit tests: no index appears in two splits, every family appears in train/val/test, no flow within the purge gap. *Deps: M1-03, M1-04 → unblocks M2-02*
- ☐ **M1-06** `nscore/features/transform.py` (`FeatureSpec`, `FlowTransformer`): fit on train only, `transform(df)` and `transform_records(list[dict])` share one code path, **fully driven by the spec file** (no hardcoded column names, so LUFlow reuses it). **Parity test**: the same flows through both paths give identical arrays. *Deps: M1-02*
- ☐ **M1-07** `nscore/drift/psi.py`: `build_reference()` (quantile bins of the top-K features) and `psi()` (epsilon-smoothed), with unit tests on synthetic shifted distributions. *Deps: M1-06*
- ☐ **M1-08** `ml/data/baseline_stats.py`: benign median and p95 per feature → `baseline_stats.json` (powers the "vs normal" explanations). Works for both schemas. *Deps: M1-05*
- ☐ **M1-09 ★** LUFlow: `ml/data/adapters/luflow.py` + `nscore/contracts/feature_spec.luflow.json` (16 CSV fields → 9 features; IPs and timestamps → metadata; **repair the corrupted time_start values**). Month-based periods. `outlier` kept as a separate label (excluded from supervised training). Train months / test months split, plus a **recent benign window** per test month for recalibration. *Deps: M1-01, M1-06 → unblocks M2-11*

- ❑ **M1-10** Replay extracts for M5 in `data/replay/`, as CSVs with metadata and ground truth, $\leq 10k$ flows each: 2018 test slices (benign background, brute-force burst, DDoS-HOIC, Ares botnet) and **LUFlow later-month slices** (mixed benign / malicious / outlier). *Deps: M1-05, M1-09*
- ❑ **M1-11** Data section of `docs/model_card.md`: both datasets and why we chose them, versions, cleaning counts, the WebAttack low-support decision, known dataset issues (Liu/Engelen et al.), split rationale, LUFlow label meanings. *Deps: M1-04, M1-05, M1-09*

M2 — ML Modeling & MLOps

Status (branch `track/M2`): M2-01 to M2-12 done; the final model card section (M2-13) is filled from the results. **The design changed after measurement:** a benign-only IsolationForest scored ~ 0 recall on CIC flows, so it is *optional* (LUFlow only) and novelty comes from the supervised forest's generalisation plus **family-head confidence** (ADR-1, `docs/experiments.md`). Tasks below keep their original wording; the notes in italics say what was actually done. Azure steps (M2-12) are implemented and tested against a fake client; **they still need your Azure subscription to run for real.**

- [x] **M2-01** ★ `scripts/make_mock_bundle.py`: a tiny bundle (RF + IF trained on random data with the feature names from `feature_spec.json`, or the fixture feature names if the spec isn't merged yet), with a valid `manifest.json` and sha256s. *Deps: — → unblocks M3-03*
- [x] **M2-02** `ml/evaluate/metrics.py`: an evaluation harness producing an `EvaluationReport` (per-class P/R/F1/FPR/AUC, macro-F1, binary ROC/PR-AUC, confusion matrix PNG + JSON). Then a **baseline** `rf_binary` on the P1 split. *Deps: M1-05, M1-06*
- [x] **M2-03** Imbalance study (*no measurable difference in-distribution; class weights kept; shortcut-feature ablation added*): `class_weight="balanced_subsample"` vs SMOTE (train only, inside the CV folds via an imblearn Pipeline, on down-sampled benign). Before/after table in `docs/experiments.md`. Pick one. *Deps: M2-02*
- [x] **M2-04** Hyperparameter search (*scored on tool-holdout recall because in-distribution is saturated; confirmed over 3 seeds*; `RandomizedSearchCV` with time-blocked / grouped CV, not plain k-fold). Set a time budget and log it. *Deps: M2-03*
- [x] **M2-05** `rf_multiclass` (*its confidence is the novelty signal*) on attack flows only (families per M1-04's Rare decision). It must expose `predict_proba` for expected severity. Per-class report. *Deps: M2-02*
- [x] **M2-06** `iforest` trained on **benign train flows only** (*result: ROC-AUC 0.70-0.87, ~ 0 recall at a usable FPR on CIC; now optional, kept for LUFlow*). Save the benign val score distribution for percentile mapping. *Deps: M1-06*
- [x] **M2-07** Threshold calibration (`tau_binary` at a 0.1% benign-FPR budget, `tau_family` from familiar validation attacks; *operating curve ships in the bundle*): original: `tau_binary` at benign FPR $\leq 1\%$ on val, `tau_anomaly` = p99.5 of benign val. Reliability plot. Write `thresholds.json`. Check `nscore/detection/fusion.py` against real numbers. *Deps: M2-04, M2-06*
- [x] **M2-08** ★ **Headline result: LOAO experiment** (`ml/train/holdouts.py`: *families + tools, 4 budgets, 3 seeds, test split*). For each 2018 family, retrain without it and record recall on it for RF-only vs fusion, at the calibrated FPR. Outputs `loao` entries in the report plus `loao.png`. Also the **HOIC tool holdout** (P2b) → `external`. *Deps: M2-07, M1-05*
- [x] **M2-09** SHAP (`nscore/detection/explain.py`; *~ 50 -110 ms/flow, so explain a few flows per incident*): `TreeExplainer` for `rf_binary` and `iforest`, a top-k local explanation function (raw value + shap + benign median), global importance, and a latency benchmark (< 50 ms/flow target). *Deps: M2-07, M1-08*

- [x] **M2-10** `nscore/bundle/` : packager (manifest + sha256) and `load_bundle("local:...")` with hash verification, supporting `family_head = False`. Build **bundle v1 (2018, full)** and **demo-holdout-botnet**. *Deps: M2-08, M2-09, M1-07 → unblocks M3 real scoring (I1)*
- [x] **M2-11 ★ LUFlow real-world study (P3 + P3b)**. Same pipeline with `feature_spec.lufLOW.json`, binary RF + IF only. Train on the earliest month(s) and test month by month (recall, FPR, PSI), plus the share of `outlier` flows flagged as novel by IF vs RF. Then refit IF + thresholds on each test month's recent benign window and re-test. Package `netsentinel-lufLOW` and `netsentinel-lufLOW-recal`. Results go into `external`, plus a month-by-month chart. *Deps: M2-07, M2-10, M1-09*
- [x] **M2-12** Azure ML (*implemented and tested with a fake client; run `scripts/azure_register.py` once your workspace exists*): create the workspace, register all four bundles with tags, implement `load_bundle("azureml:...")` with a cache fallback, and **prove it works by pulling into a clean environment**. *Deps: M2-11*
- [x] **M2-13** Model card (`docs/model_card.md`): metrics, LOAO, operating point, **real-world results (LUFlow)**, a **drift section with measured numbers**, limitations. Done with M1 (data section) and M5 (Q&A wording). *Deps: M2-08, M2-11*

M3 — Backend & Platform

- ☐ **M3-01 ★** FastAPI skeleton with **mock mode** (`NS MOCK=1`): every endpoint in 03 §4.1 returns the matching fixture, typed with the contract models. `/docs` works. *Deps: — → unblocks M4*
- ☐ **M3-02** SQLite schema (03 §4.5), WAL mode, repository layer, `scripts/init_db.py`. *Deps: —*
- ☐ **M3-03** Load the bundle at startup via `load_bundle(MODEL_REF)` (mock bundle until v1 exists). Real `/health`, `/v1/model` (with `feature_schema`, `family_head`), `/v1/model/evaluation`. *Deps: M2-01*
- ☐ **M3-04** `POST /v1/flows` : validate against the **loaded bundle's** spec → `FlowTransformer` → RF + IF → `fuse` → family + probabilities (or `malicious` if there's no family head) → SHAP for non-benign → persist. Batches of up to 500. Record latency per flow. *Deps: M3-02, M3-03, M1-06*
- ☐ **M3-05** Incident correlator (03 §4.3) with unit tests: windowing, key separation, no merging into resolved or dismissed incidents, running mean of severity. *Deps: M3-04*
- ☐ **M3-06** Risk and MITRE enrichment via `nscore.contracts.policy` (`expected_severity`, `risk_score`, `risk_level`, `mitre_for`). Recompute on every upsert. *Deps: M3-05*
- ☐ **M3-07** `GET /v1/incidents` (filter/sort by risk/paginate), `GET /v1/incidents/{id}`, `POST /v1/incidents/{id}/actions` with an audit row and status transitions (stamp `acknowledged_at` for MTTA). *Deps: M3-05*
- ☐ **M3-08** Drift monitor service (rolling window → `nscore.drift.psi`, snapshots tagged with model version) and `/v1/drift`. `/v1/metrics` (throughput, p50/p95, confirmed precision, FP rate, MTTA). *Deps: M3-04, M1-07*
- ☐ **M3-09** Wire in M5's brief service: `GET /v1/incidents/{id}/brief` (lazy, cached, timeout → template). `POST /v1/admin/reload-model` (switch bundle, reset the drift window; must work across schemas: 2018 ↔ LUFlow). *Deps: M3-07, M5-03*
- ☐ **M3-10** Hardening: API key, `x-Analyst`, admin key, structured JSON logs with request ID, 422 messages that name missing features, CORS. *Deps: M3-07*
- ☐ **M3-11** Tests (`api/tests/`): every endpoint, happy path and errors (malformed flow, wrong-schema flow, missing model), against the mock bundle. Make the CI checks required. `infra/docker-compose.yml` (api + dashboard, one command). *Deps: M3-07*
- ☐ **M3-12 (stretch)** Deploy the API and dashboard to Azure Container Apps. *Deps: M3-11*

M4 — SOC Console

- ☐ **M4-01 ★** Streamlit skeleton + `dashboard/api_client.py` typed with the contract models. Two data sources: `NS_API_URL` (live or mock API) and an **offline mode** that reads the fixtures directly. You can start this today. *Deps: —*
- ☐ **M4-02** Visual system: HIGH / MEDIUM / LOW colours (red / amber / grey), `Known attack` / `Novel anomaly` chips, status pills, dark SOC theme, `dashboard/theme.py`. *Deps: M4-01*
- ☐ **M4-03 Live Queue** page: sorted by risk score, filters, header counters, loaded-bundle badge (2018 / LUFlow), 3 s auto-refresh (`st.fragment(run_every=...)`), new-row highlight, pagination for 500+ incidents. *Deps: M4-02*
- ☐ **M4-04 Incident Detail: risk breakdown** (confidence × severity × burst = score), SHAP bar chart, **"vs normal" table**, MITRE badge linking to `attack.mitre.org`, metadata, brief panel (loading / LLM / template label), action buttons with a note, action timeline. Hide family widgets when `family_head` is false. *Deps: M4-03*
- ☐ **M4-05** Analyst sign-in (name + role) in session state, sent as `X-Analyst`. *Deps: M4-04*
- ☐ **M4-06 Model & Evaluation** page: version and registry ref, per-class table, confusion-matrix heatmap, **LOAO grouped bar chart** (RF-only vs fusion), **Real-world panel** (LUFlow month-by-month recall / FPR / PSI, before vs after recalibration, from `external`), operating FPR, limitations. *Deps: M4-02*
- ☐ **M4-07 Drift & Health** page: PSI bars with 0.10/0.25 reference lines, status banner, throughput and latency tiles. *Deps: M4-02*
- ☐ **M4-08 Analyst Metrics** page: confirmed precision, FP dismiss rate, MTTA, actions per analyst. *Deps: M4-02*
- ☐ **M4-09** Switch from the mock API to the real one at I1, and fix any contract gaps (raise a contract PR, don't hack around it). *Deps: M3-07*
- ☐ **M4-10** Polish: empty, error and loading states, API-down banner, a screenshot set for the deck. *Deps: M4-09*

M5 — Security, GenAI & Demo

- ☐ **M5-01** Threat model: turn the team's **Risk Scoring Pipeline** doc into `docs/threat_model.md`, with the written reason for every weight in `policy.SEVERITY` and validation of `MITRE_MAP` (especially Infiltration). Write one "suggested next step" playbook line per family (used by the template brief). *Deps: —*
- ☐ **M5-02** Provision Azure OpenAI and a gpt-4o-mini class deployment. Fill in `.env.example` keys and document the setup in `docs/azure_setup.md` (shared with M2's Azure ML notes). *Deps: —*
- ☐ **M5-03 ★** `api/app/services/brief.py`: `generate_brief(incident: IncidentDetail) -> Brief`. Grounded prompt (03 §4.7), confidence-band hedging, novel-anomaly and `Malicious` wording, 8 s timeout, **deterministic template fallback**. Unit tests with a stubbed client. Built against `fixtures/incident_detail.json`, so it doesn't need the API. *Deps: M5-01, M5-02 → unblocks M3-09*
- ☐ **M5-04** Brief evaluation: 10 varied incidents (each family, low and high confidence, novel, LUFlow malicious). Check for invented facts and correct hedging. Results table in `docs/brief_eval.md`. *Deps: M5-03*
- ☐ **M5-05** `replay/replay.py`: reads replay CSVs → batches → `POST /v1/f1ows` with pacing (`--speed`), scenario files `replay/scenarios/*.yaml` (each scenario names the bundle it needs), and live ground-truth scoring printed to the console. Built against the mock API first. *Deps: M3-01*
- ☐ **M5-06** Demo design: write `docs/demo_script.md` (the 03 §6 storyline with timings, exact commands, bundle reloads and expected screens), including act 5 (LUFlow later month → drift alert → LUFlow-recal → recovery). *Deps: M5-05, M1-10*

- ☐ **M5-07** (*stretch*) Own-lab capture: two VMs on a host-only network, a slow nmap scan against **our own VM only**, fixed CICFlowMeter → CSV → replay. Document the safety and legality notes. *Deps: M5-05*
 - ☐ **M5-08** Fix the **Round 1 doc** using 02 A1–A10 (dataset is now 2018 + LUFlow), and build the **pitch deck. Q&A prep sheet** covering: "it's supervised, how does it catch novel attacks?", "what if the model is wrong?", "isn't this lab data?" (→ LUFlow), "why 2018?" (→ ADR-8), "how is risk computed?" (→ ADR-9, worked examples), "1% FPR at scale?", "drift?" (→ measured on LUFlow). *Deps: —, then update after I2*
 - ☐ **M5-09** Run 2 timed end-to-end dry runs and record the fallback video. Cross-check every deck claim against what's actually built. *Deps: I2*
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Working agreements

- **Branches:** `M3/04-flows-endpoint` → PR to `main`. Small PRs, at least one reviewer, CI must pass.
 - **Contracts:** any change under `nscore/contracts/` needs approval from M3 plus one consumer, a bump to `CONTRACT_VERSION`, and regenerated fixtures (`python -m nscore.contracts.fixtures.make_fixtures`).
 - **Data and secrets:** never commit `data/` , `artifacts/` or `.env` . Gitleaks runs in CI.
 - **Honesty rule:** no number goes in the deck unless the repo reproduces it.
 - **Stuck for more than 2 hours on a dependency?** Build against the fixture or mock, and flag it in the track issue.
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Source: docs/04_tasks.md · github.com/Saroy-ctrl/netsentinel